MATH329 · OPERATIONS RESEARCH

Midterm 471 Solutions

60 minutes

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01 Solution Q1: True / False

#	Answer	Reason
1	**F**	The model's parameters are known. The unknowns are the variables.
2	**T**	This is the definition of a BFS.
3	**T**	This is the definition of degeneracy.
4	**F**	If all the ratios are negative or infinite, no variable can leave and the solution is unbounded.
5	**F**	The solution must satisfy the constraints.

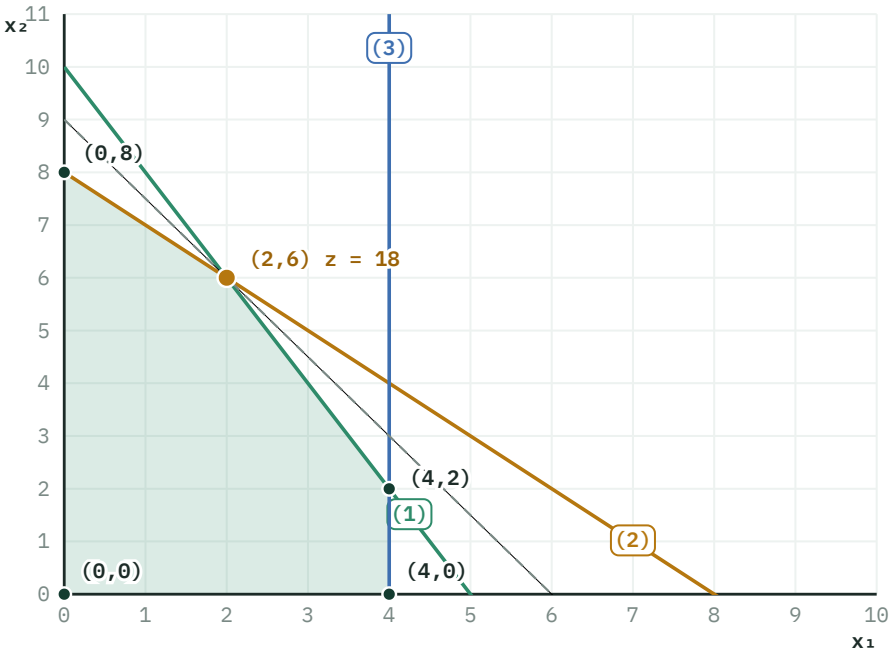
02 Solution Q2: graphical

1 Prepare the constraints

(1) $2x_1 + x_2 = 10$ points (0, 10) and (5, 0)

(2) $x_1 + x_2 = 8$ points (0, 8) and (8, 0)

(3) $-x_1 \geq -4 \times (-1) \rightarrow x_1 \leq 4$ vertical line at $x_1 = 4$



2 The corner points

Corner	Where it comes from	$z = 3x_1 + 2x_2$
(0, 0)	origin	0
(4, 0)	(3) with the x_1 axis	12
(4, 2)	(1) with (3)	16
(2, 6)	(1) with (2)	**18**
(0, 8)	(2) with the x_2 axis	16

(1) − (2): $x_1 = 2$ $\quad 2 + x_2 = 8 \quad \rightarrow \quad x_2 = 6$

Objective Value = 18 at $x_1 = 2$ and $x_2 = 6$

03 Solution Q3: standard form and simplex

a Standard form

$$\text{Maximize } Z = 2x_1 + 3x_2 + 0s_1 + 0s_2$$

$$\text{S.to} \quad x_1 + 2x_2 + s_1 = 16$$

$$2x_1 + x_2 + s_2 = 18$$

$$x_1, x_2, s_1, s_2 \geq 0$$

b Simplex

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
Z	−2	−3	0	0	0	
s_1	1	**2**	1	0	16	8
s_2	2	1	0	1	18	18
Z	$-\frac{1}{2}$	0	$\frac{3}{2}$	0	24	
x_2	$\frac{1}{2}$	1	$\frac{1}{2}$	0	8	16
s_2	**$\frac{3}{2}$**	0	$-\frac{1}{2}$	1	10	$\frac{20}{3}$
Z	0	0	$\frac{4}{3}$	$\frac{1}{3}$	$\frac{82}{3}$	
x_2	0	1	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{14}{3}$	
x_1	1	0	$-\frac{1}{3}$	$\frac{2}{3}$	$\frac{20}{3}$	

Iteration 1: x_2 enters because -3 is the most negative. The ratios are $\frac{16}{2} = 8$ and $\frac{18}{1} = 18$, so s_1 leaves and the pivot is 2.

Iteration 2: $-\frac{1}{2}$ remains under x_1, so it enters. The ratios are $\frac{8}{\frac{1}{2}} = 16$ and $\frac{10}{\frac{3}{2}} = 6.67$, so s_2 leaves and the pivot is $\frac{3}{2}$.

Optimum $z = \frac{82}{3} = 27.33$ at $x_1 = \frac{20}{3} = 6.67$ and $x_2 = \frac{14}{3} = 4.67$

Check: $z = 2\left(\frac{20}{3}\right) + 3\left(\frac{14}{3}\right) = \frac{40}{3} + \frac{42}{3} = \frac{82}{3}$

The key writes decimals: 1.33, 0.33, 27.33. Its marking: three marks for each table and three for the optimum, with half a mark off for each arithmetic slip.